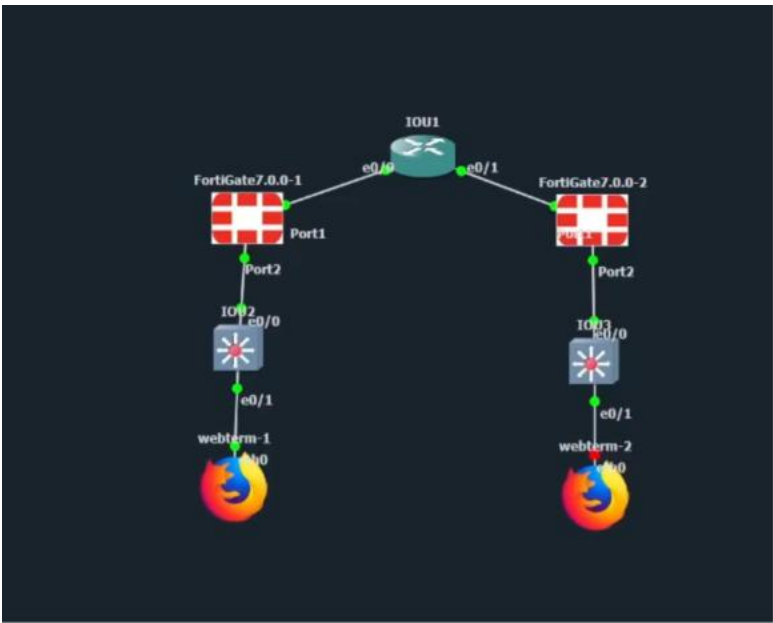


1. Objetivo

Configurar una VPN Site-to-Site entre dos FortiGates usando IPSec, permitiendo comunicación cifrada entre las redes LAN de dos sitios a través de un router ISP que simula internet. Se verifica la conectividad y el cifrado del tráfico mediante traceroute entre ambas LANs.

2. Topología de Red



```
[PC1]--[SWA]--[FGT-A Port2]  [FGT-A Port1]--[ISP]--[FGT-B Port1]  [FGT-B Port2]--[
[SWB]--[PC2]
                10.11.85.1/26   10.11.85.65/30       10.11.85.70/30   10.11.85.129/26
                        .66   ISP   .69
```

Roles de cada dispositivo

Dispositivo	Rol	Descripción
FortiGate-A	Peer VPN Sitio A	Termina el túnel IPSec del sitio A, rutea tráfico cifrado
FortiGate-B	Peer VPN Sitio B	Termina el túnel IPSec del sitio B, rutea tráfico cifrado
ISP	Tránsito	Simula internet. Solo rutea, no participa en el túnel

VPN Site-to-Site

PC1	Host Sitio A	Host en la LAN A que se comunica con PC2 a través del túnel
PC2	Host Sitio B	Host en la LAN B que se comunica con PC1 a través del túnel

Direccionamiento IP

Dispositivo	Interfaz	Dirección IP	Descripción
FortiGate-A	port1	10.11.85.65/30	WAN → ISP
FortiGate-A	port2	10.11.85.1/26	LAN → SWA → PC1
ISP	e0/0	10.11.85.66/30	Hacia FortiGate-A
ISP	e0/1	10.11.85.69/30	Hacia FortiGate-B
FortiGate-B	port1	10.11.85.70/30	WAN → ISP
FortiGate-B	port2	10.11.85.129/26	LAN → SWB → PC2
PC1	e0	10.11.85.2/26	GW: 10.11.85.1
PC2	e0	10.11.85.130/26	GW: 10.11.85.129

3. Parámetros VPN

Phase 1 — IKE

Parámetro	Valor
Versión IKE	1
Modo	Main (ID protection)
Autenticación	Pre-Shared Key
PSK	ITLA2024
Cifrado	AES-256
Hash	SHA-256
Grupo DH	14 (2048-bit)
Lifetime	86400 segundos (24 horas)

Phase 2 — IPSec

Parámetro	Valor
Cifrado	AES-256
Hash	SHA-256
PFS	Habilitado
Lifetime	43200 segundos
Subred local FGT-A	10.11.85.0/26
Subred remota FGT-A	10.11.85.128/26
Subred local FGT-B	10.11.85.128/26
Subred remota FGT-B	10.11.85.0/26

4. Configuración

ISP — Cisco IOU (CLI)

```
enable
configure terminal
hostname ISP
!
interface Ethernet0/0
 ip address 10.11.85.66 255.255.255.252
 no shutdown
!
interface Ethernet0/1
 ip address 10.11.85.69 255.255.255.252
 no shutdown
!
ip route 10.11.85.0 255.255.255.192 10.11.85.65
ip route 10.11.85.128 255.255.255.192 10.11.85.70
!
end
write memory
```

FortiGate-A — Interfaces (CLI)

```
config system interface
 edit port1
  set mode static
  set ip 10.11.85.65 255.255.255.252
  set allowaccess ping https ssh http
 next
 edit port2
  set mode static
  set ip 10.11.85.1 255.255.255.192
  set allowaccess ping https ssh http
 next
end
!
config router static
 edit 1
  set gateway 10.11.85.66
  set device port1
 next
end
```

FortiGate-B — Interfaces (CLI)

```
config system interface
 edit port1
  set mode static
  set ip 10.11.85.70 255.255.255.252
  set allowaccess ping https ssh http
 next
 edit port2
  set mode static
```

```
set ip 10.11.85.129 255.255.255.192
set allowaccess ping https ssh http
next
end
!
config router static
edit 1
set gateway 10.11.85.69
set device port1
next
end
```

FortiGate-A — VPN (GUI: VPN → IPsec Wizard)

The screenshot shows the FortiGate web interface at the URL `http://10.11.85.1/ng/vpn/ipsec/wizard`. The left sidebar menu is expanded to the 'VPN' section, with 'IPsec Wizard' selected. The main content area displays the 'VPN Creation Wizard' progress bar, which is complete for all steps: VPN Setup, Authentication, Policy & Routing, and Review Settings. A green message box states 'The VPN has been set up'. Below this, the 'Object Summary' table lists the configured VPN parameters.

Object Summary	
Phase 1 interface	✓ VPNFORTIA Edit
Local address group	✓ VPNFORTIA_local Edit
Remote address group	✓ VPNFORTIA_remote Edit
Phase 2 interface	✓ VPNFORTIA
Static route	✓ 2 Edit
Blackhole route	✓ 3 Edit
Local to remote policies	✓ vpn_VPNFORTIA_local_0 (1)
Remote to local policies	✓ vpn_VPNFORTIA_remote_0 (2)

At the bottom of the summary section are two buttons: 'Add Another' and 'Show Tunnel List'.

Campo	Valor
Name	VPN-A-a-B
Template	Site to Site
Remote IP	10.11.85.70
Interface	port1

VPN Site-to-Site

Pre-shared Key	ITLA2026
Local interface	port2
Local subnet	10.11.85.0/26
Remote subnet	10.11.85.128/26

FortiGate-B — VPN (GUI: VPN → IPsec Wizard)

FortiGate - FortiGate-VM64-KVM — Mozilla Firefox

FortiGate - FortiGate-VM64-KVM

Not Secure http://10.11.85.1/ng/vpn/ipsec/wizard

FortiGate-VM64-KVM

VPN Creation Wizard

VPN Setup Authentication Policy & Routing Review Settings

The following settings should be reviewed prior to creating the VPN.

Object Summary

Phase 1 interface	VPN-SITE-A-SITEB
Local address group	VPN-SITE-A-SITEB_local
Remote address group	VPN-SITE-A-SITEB_remote
Phase 2 interface	VPN-SITE-A-SITEB
Static route	static
Blackhole route	static
Local to remote policies	vpn_VPN-SITE-A-SITEB_local
Remote to local policies	vpn_VPN-SITE-A-SITEB_remote

< Back Create Cancel

FORTINET v7.0.0

Campo	Valor
Name	VPN-B-a-A
Template	Site to Site
Remote IP	10.11.85.65
Interface	port1

VPN Site-to-Site

Pre-shared Key	ITLA2026
Local interface	port2
Local subnet	10.11.85.128/26
Remote subnet	10.11.85.0/26

VPCS

```
! PC1
ip 10.11.85.2 255.255.255.192 10.11.85.1
save

! PC2
ip 10.11.85.130 255.255.255.192 10.11.85.129
save
```

5. Verificación

[Agregar aquí capturas de pantalla de los comandos de verificación]

En FortiGate CLI

```
! Ver estado del túnel IPsec
get vpn ipsec tunnel summary

FortiGate-VM64-KVM # get vpn ipsec tunnel summary
'VPNFORTIA' 10.11.85.70:0 selectors(total,up): 1/1 rx(pkt,err): 8/0 tx(pkt,err): 8/2
FortiGate-VM64-KVM # get vpn ipsec tunnel summary
```

Desde PC1 — Traceroute a PC2

```
ping 10.11.85.130
trace 10.11.85.130
64 bytes from 10.11.85.130: icmp_seq=24 ttl=64 time=40.2 ms
^C
--- 10.11.85.130 ping statistics ---
31 packets transmitted, 31 received, 0% packet loss, time 31761ms
rtt min/avg/max/mdev = 4.135/11.307/54.223/8.971 ms
root@webterm-1:~# traceroute 10.11.85.130
traceroute to 10.11.85.130 (10.11.85.130), 30 hops max, 60 byte packets
 1 10.11.85.1 (10.11.85.1) 29.280 ms 28.761 ms 25.636 ms
 2 10.11.85.70 (10.11.85.70) 73.802 ms 73.570 ms 73.483 ms
 3 10.11.85.130 (10.11.85.130) 44.717 ms 53.040 ms 56.400 ms
root@webterm-1:~#
```

Desde PC2 — Traceroute a PC1

```
ping 10.11.85.2
trace 10.11.85.2
rtt min/avg/max/mdev = 2.436/17.375/115.370/30.491 ms
root@webterm-2:~# ping 10.11.85.2
PING 10.11.85.2 (10.11.85.2) 56(84) bytes of data:
64 bytes from 10.11.85.2: icmp_seq=1 ttl=62 time=58.8 ms
64 bytes from 10.11.85.2: icmp_seq=2 ttl=62 time=9.63 ms
64 bytes from 10.11.85.2: icmp_seq=3 ttl=62 time=4.94 ms
64 bytes from 10.11.85.2: icmp_seq=4 ttl=62 time=7.58 ms
^C
traceroute to 10.11.85.2 (10.11.85.2), 30 hops max, 60 byte packets
 1 10.11.85.129 (10.11.85.129) 97.331 ms 91.545 ms 84.745 ms
 2 10.11.85.65 (10.11.85.65) 135.002 ms 134.913 ms 127.782 ms
 3 10.11.85.2 (10.11.85.2) 97.647 ms 97.661 ms 97.668 ms
root@webterm-2:~#
```


VPN Site-to-Site

FortiGate-VM64-KVM

Dashboard Network Policy & Objects Security Profiles VPN Overlay Controller VPN IPsec Tunnels IPsec Wizard IPsec Tunnel Template SSL-VPN Portals SSL-VPN Settings SSL-VPN Clients VPN Location Map User & Authentication System Security Fabric Log & Report

Not Secure http://10.11.85.1/ng/vpn/ipsec

admin

+ Create New Edit Delete Search

Tunnel	Interface Binding	Status	Ref.
Site to Site - FortiGate 1			
VPNFORTIA	port1	Up	4

FortiGate - FortiGate-VM64-KVM

Dashboard Network Policy & Objects Security Profiles VPN Overlay Controller VPN IPsec Tunnels IPsec Wizard IPsec Tunnel Template SSL-VPN Portals SSL-VPN Settings SSL-VPN Clients VPN Location Map User & Authentication System Security Fabric Log & Report

Not Secure http://10.11.85.129/ng/vpn/ipsec

admin

+ Create New Edit Delete Search

Tunnel	Interface Binding	Status	Ref.
Site to Site - FortiGate 1			
VPNFORTIB	port1	Up	4

FortiNET v7.0.0 0 Security Rating Issues